

Homework_7

```
zsh: command not found: -zsh
Mohana_Alla 🙄🙄🙄 ~ % wget http://hgdownload.cse.ucsc.edu/goldenpath/hg19/bigZips/hg19.fa.gz
--2025-04-10 15:02:37-- http://hgdownload.cse.ucsc.edu/goldenpath/hg19/bigZips/hg19.fa.gz
Resolving hgdownload.cse.ucsc.edu (hgdownload.cse.ucsc.edu)... 128.114.119.163
Connecting to hgdownload.cse.ucsc.edu (hgdownload.cse.ucsc.edu)|128.114.119.163|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 948731419 (905M) [application/x-gzip]
Saving to: 'hg19.fa.gz'

hg19.fa.gz                               99%[=====
===== ] 904.35hg19.fa.gz
    100%[=====>] 904.
78M  1.61MB/s   in 6m 25s

2025-04-10 15:09:01 (2.35 MB/s) - 'hg19.fa.gz' saved [948731419/948731419]
```

```
Mohana_Alla 🙄🙄🙄 ~ % bedtools getfasta -fi hg19.fa -bed MCF7_POLR2A_peaks.narrowPeak -fo MCF7_POLR2
A_peaks.narrowPeak_Sequences.fasta
Mohana_Alla 🙄🙄🙄 ~ %
```

MEME-ChIP
Motif Analysis of Large Nucleotide Datasets

MEME Suite 5.5.7

Queue status: OPEN
Jobs running: 14
Jobs waiting to run: 9

► Motif Discovery

► Motif Enrichment

► Motif Scanning

► Motif Comparison

► Gene Regulation

► Utilities

► Manual

► Guides & Tutorials

► Sample Outputs

► File Format Reference

► Databases

► Download & Install

► Help

► Alternate Servers

► Authors & Citing

► Recent Jobs

Previous version 5.5.6

Please wait. Your MEME-ChIP job is now queued awaiting available resources. You may bookmark this page or use the **Recent Jobs** menu at the left to access your job's results.

Job Details ▼

Submitted	4/10/2025, 11:48:06 PM
Expires	4/14/2025, 11:48:06 PM
(Primary) Sequences	A set of 33319 DNA sequences, between 88 and 9733 in length (average length 226.3), from the file <code>MCF7_POLR2A_peaks.narrowPeak_Sequences.fasta</code> .
Motifs	A set of 1956 motifs, between 5 and 35 in width (average width 12.5), from the database <code>JASPAR CORE (2022)</code> .
Motif Width	MEME and STREME will find motifs that are between 6 and 15 wide (inclusive).
Background	A 2-order background model generated from the supplied sequences.
Parse Genomic Coordinates	Motif sites will be output in genomic coordinates if possible.
Discovery Mode	Classic: optimizes the E-value of the motif information content
MEME Site Distribution	MEME will only find motifs with zero or one occurrence (of a contributing motif site) per sequence.
MEME Motif Count	MEME will search for 3 motifs.
STREME <i>p</i> -value Threshold	STREME will search for motifs with a <i>p</i> -value ≤ 0.05 .
STREME Site Positional Distribution Plots	Sequences are aligned on their centers.
CentriMo Match Score	CentriMo will only use motif sites if they have a match score ≥ 5 .
CentriMo <i>E</i> -value Threshold	CentriMo will only report motifs which have a region enrichment <i>E</i> -value ≤ 10 .
CentriMo Local	CentriMo will only find central enriched regions.
CentriMo Store IDs	CentriMo will store the sequence identifiers and their mapping to the best region of all significant motifs.

MEME-CHIP

Motif Analysis of Large Nucleotide Datasets

For further information on how to interpret these results please access <https://meme-suite.org/meme/doc/meme-chip-output-format.html>.
To get a copy of the MEME software please access <https://meme-suite.org>.

If you use MEME-CHIP in your research, please cite the following paper:
Philip Machanick and Timothy L. Bailey, "MEME-CHIP: motif analysis of large DNA datasets", *Bioinformatics* 27(12):1696-1697, 2011. [\[full text\]](#)

[MOTIFS](#) | [PROGRAMS](#) | [INPUT FILES](#) | [PROGRAM INFORMATION](#) | [SUMMARY IN TSV FORMAT](#) | [MOTIFS IN MEME TEXT FORMAT](#)

MOTIFS

The significant motifs (E-value ≤ 0.05) found by the programs MEME, STREME and CentriMo; clustered by similarity and ordered by E-value.					
Expand All Clusters Collapse All Clusters					
Motif Found Reverse Complement ↔	Discovery/Enrichment Program MEME	E-value 1.3e-033	Known or Similar Motifs DOF3.6 (MA1274.1) FoxM1 (MA1871.1) DOF5.8 (MA1267.1)	Distribution Not Centrally Enriched	SpaMo & FIMO - Motif Spacing Analysis - Motif Sites in GFF3
Motif Found Reverse Complement ↔	Discovery/Enrichment Program STREME	E-value 1.6e-014	Known or Similar Motifs ZNF460 (MA1596.1) SP4 (MA0685.2) SPI1 (MA0079.5)	Distribution Not Centrally Enriched	SpaMo & FIMO - Motif Spacing Analysis - Motif Sites in GFF3
Motif Found Reverse Complement ↔	Discovery/Enrichment Program STREME	E-value 1.9e-014	Known or Similar Motifs YY2 (MA0748.2) ERF13 (MA1004.1) ERF096 (MA0998.1)	Distribution Not Centrally Enriched	SpaMo & FIMO - Motif Spacing Analysis - Motif Sites in GFF3

Motif Found Reverse Complement ↔ Show 1 More ⓘ	Discovery/Enrichment Program STREME	E-value 2.1e-009	Known or Similar Motifs	Distribution Not Centrally Enriched	SpaMo & FIMO - Motif Spacing Analysis - Motif Sites in GFF3
Motif Found Reverse Complement ↔	Discovery/Enrichment Program STREME	E-value 2.3e-007	Known or Similar Motifs Zn000014020267 (MA1817.1) Zn000014049264 (MA1833.1) LOB (MA2022.1)	Distribution Not Centrally Enriched	SpaMo & FIMO - Motif Spacing Analysis - Motif Sites in GFF3
Motif Found Reverse Complement ↔	Discovery/Enrichment Program STREME	E-value 5.7e-007	Known or Similar Motifs	Distribution Not Centrally Enriched	SpaMo & FIMO - Motif Spacing Analysis - Motif Sites in GFF3
Motif Found Reverse Complement ↔	Discovery/Enrichment Program STREME	E-value 6.8e-007	Known or Similar Motifs Nrf1 (MA0506.2) TSAR2 (MA1412.1)	Distribution Not Centrally Enriched	SpaMo & FIMO - Motif Spacing Analysis - Motif Sites in GFF3
Motif Found Reverse Complement ↔	Discovery/Enrichment Program STREME	E-value 1.1e-006	Known or Similar Motifs ZBTB6 (MA1581.1) RAV1 (MA0583.1)	Distribution Not Centrally Enriched	SpaMo & FIMO - Motif Spacing Analysis - Motif Sites in GFF3
Motif Found	Discovery/Enrichment Program	E-value	Known or Similar Motifs	Distribution	SpaMo & FIMO

meme-suite.org					
Assignments: BIOINFORMATICS: ADV BIO APPS - 1	MEME-CHIP Results	ChatGPT	bioinformatics jobs - Google Search		
Motif Found Reverse Complement ⇌	Discovery/Enrichment Program STREME	E-value 4.6e-006	Known or Similar Motifs PAX9 (MA0781.1) PAX1 (MA0729.1) GF11 (MA0038.2)	Distribution Not Centrally Enriched	SpaMo & FIMO Motif Spacing Analysis Motif Sites in GFF3
Motif Found Reverse Complement ⇌ Show 1 More ⓘ	Discovery/Enrichment Program STREME	E-value 3.7e-005	Known or Similar Motifs (None listed)	Distribution Not Centrally Enriched	SpaMo & FIMO Motif Spacing Analysis Motif Sites in GFF3
Motif Found Reverse Complement ⇌	Discovery/Enrichment Program STREME	E-value 2.6e-004	Known or Similar Motifs ELK4 (MA0076.2) ELK3 (MA0759.2) FEV (MA0156.3)	Distribution Not Centrally Enriched	SpaMo & FIMO Motif Spacing Analysis Motif Sites in GFF3
Motif Found Reverse Complement ⇌	Discovery/Enrichment Program MEME	E-value 9.3e-004	Known or Similar Motifs GRF4 (MA1815.1) ZNF92 (MA1721.1) ERF::NHLH1 (MA1938.1)	Distribution Not Centrally Enriched	SpaMo & FIMO Motif Spacing Analysis Motif Sites in GFF3
Motif Found Reverse Complement ⇌	Discovery/Enrichment Program STREME	E-value 3.6e-003	Known or Similar Motifs TCP17 (MA1290.1) TCP13 (MA1282.1) kn (MA1841.1)	Distribution Not Centrally Enriched	SpaMo & FIMO Motif Spacing Analysis Motif Sites in GFF3
Motif Found 	Discovery/Enrichment Program (None listed)	E-value (None listed)	Known or Similar Motifs (None listed)	Distribution (None listed)	SpaMo & FIMO (None listed)

meme-suite.org					
Assignments: BIOINFORMATICS: ADV BIO APPS - 1	MEME-CHIP Results	ChatGPT	bioinformatics jobs - Google Search		
Motif Found Reverse Complement ⇌	Discovery/Enrichment Program STREME	E-value 4.8e-003	Known or Similar Motifs ZNF460 (MA1596.1)	Distribution Not Centrally Enriched	SpaMo & FIMO Motif Spacing Analysis Motif Sites in GFF3
Motif Found Reverse Complement ⇌	Discovery/Enrichment Program STREME	E-value 1.5e-002	Known or Similar Motifs unc-62 (MA0918.1)	Distribution Not Centrally Enriched	SpaMo & FIMO Motif Spacing Analysis Motif Sites in GFF3

PROGRAMS

Command	Running Time	Status	Outputs
getsize ./MCF7_POLR2A_peaks.narrowPeak_Sequences.fasta 1> \$metrics	0.09s	Success	
fasta-most -min 50 < ./MCF7_POLR2A_peaks.narrowPeak_Sequences.fasta 1> \$metrics	0.21s	Success	
fasta-center -dna -len 100 < ./MCF7_POLR2A_peaks.narrowPeak_Sequences.fasta 1> ./seqs-centered	0.39s	Success	seqs-centered
fasta-get-markov -nostatus -nosummary -dna -m 2 ./MCF7_POLR2A_peaks.narrowPeak_Sequences.fasta ./background	0.07s	Success	Background
meme ./seqs-centered -oc meme_out -mod zoops -nmotifs 3 -minw 6 -maxw 15 -bfile ./background -dna -seed 0 -searchsize 100000 -time 2919 -revcomp -nostatus	19m 45.64s	Success	MEME HTML MEME text MEME XML
streme --verbosity 1 --oc streme_out -dna --seed 0 --p ./seqs-centered --minw 6 --maxw 15 --time 5685 --order 2 --thresh 0.05 --align center --totallength 4000000	59m 50.36s	Warnings	STREME HTML STREME text STREME XML STREME SEQUENCES STREME SITES
centrimo -seqlen 88 -verbosity 1 -oc centrimo_out -bfile ./background -score 5.0 -ethresh 10.0 ./MCF7_POLR2A_peaks.narrowPeak_Sequences.fasta meme_out/meme.xml streme_out/streme.xml db/motif_databases/JASPAR/JASPAR2022_CORE_non-redundant_v2.meme	16.33s	Warnings	CentriMo HTML Site Counts
tomtom -verbosity 1 -oc meme_tomtom_out -min-overlap 5 -dist pearson -eval -thresh 1 -no-ssc meme_out/meme.xml db/motif_databases/JASPAR/JASPAR2022_CORE_non-redundant_v2.meme	11.60s	Success	Tomtom HTML Tomtom TSV Tomtom XML
tomtom -verbosity 1 -oc streme_tomtom_out -min-overlap 5 -dist pearson -eval -thresh 1 -no-ssc streme_out/streme.xml db/motif_databases/JASPAR/JASPAR2022_CORE_non-redundant_v2.meme	1m 28.27s	Success	Tomtom HTML Tomtom TSV Tomtom XML
tomtom -verbosity 1 -text -thresh 0.1 -combined meme ./combined meme 1> ./motif_alignment.txt	0.46s	Success	Motif Alignment

MOTIF 1

Sequence: TTTDTWTTTTTTKTW

Type: MEME

Length: 15

Letter-probability matrix:

- Extremely high T content across the motif, particularly positions 4–13.
- Virtually no G or C; mostly A/T combinations.
- Likely represents AT-rich regulatory regions such as promoters or nucleosome-free zones.

MOTIF 2

Sequence: CTCVGCCTC

Type: STREME

Length: 9

Letter-probability matrix:

- Strong C dominance in most positions, especially 1, 3, 6, 8, and 9.
- G is highly enriched in the center (position 5).
- Suggests a palindromic C-rich motif, possibly bound by zinc finger proteins or CTCF.

MOTIF 3

Sequence: AAGATGGCGGCS

Type: STREME

Length: 12

Letter-probability matrix:

- High A presence at the start; strong G/C enrichment from positions 5 to 11.
- Suggests involvement in GC box regions.
- Likely recognized by transcription factors like SP1 or EGR1.

MOTIF 4

Sequence: CCAGCCTGGGCRACA

Type: STREME

Length: 15

Letter-probability matrix:

- Rich in C and G through the middle of the sequence.
- Resembles enhancer elements found in CpG islands.
- Suggests involvement in gene regulation in GC-dense genomic regions.

MOTIF 5

Sequence: GCGGCGGCGGCGGC

Type: STREME

Length: 14

Letter-probability matrix:

- Almost perfect GC repeats, forming a symmetrical motif.
- Very high G/C probability across all positions.
- Classic GC-box motif; a known binding site for SP1 transcription factor.

MOTIF 6

Sequence: RCCTGTARTCCCAGC

Type: STREME

Length: 15

Letter-probability matrix:

- Central TCCC cluster with upstream TGTA-like sequence.
- C- and G-rich zones suggest a complex binding landscape.
- May indicate p53 family, RUNX, or related TF binding.

MOTIF 7

Sequence: GCGCATGCGCA

Type: STREME

Length: 11

Letter-probability matrix:

- Palindromic GC-rich structure, symmetrical about the center.
- Shows perfect complementarity in many positions.
- Likely bound by factors sensitive to DNA methylation or chromatin state.

MOTIF 8

Sequence: ATCRCTTGAGCCCAG

Type: STREME

Length: 15

Letter-probability matrix:

- AT-rich start transitions into GAG and CCC core.
- Enriched in purines and pyrimidines suggesting flexibility in recognition.
- May be part of enhancer modules activated under stimulus.

MOTIF 9

Sequence: TGCAGTGAGCYRWGA

Type: STREME

Length: 15

Letter-probability matrix:

- Strong TGAG motif in the middle; classic of hormone response elements.
- Variable ends, allowing diverse TF recognition.
- Likely related to nuclear receptor pathways or cytokine signaling.

MOTIF 10

Sequence: CAGGCTGGAGTGCAG

Type: MEME

Length: 15

Letter-probability matrix:

- GGTGCA core tightly conserved; G/C-rich throughout.
- Flanked by A/C at both ends, potentially stabilizing TF-DNA binding.
- Suggests estrogen receptor binding, common in breast cancer lines like MCF-7.

For K562

```
zsh: command not found: -zsh
Mohana_Alla 🤔🤔🤔 ~ % bedtools getfasta -fi hg19.fa -bed K562_POLR2A_peaks.narrowPeak -fo K562_POLR2A_peak
s.narrowPeak_Sequences.fasta
```



MEME-ChIP

Motif Analysis of Large Nucleotide Datasets

Please wait. Your MEME-ChIP job is now queued awaiting available resources. You may bookmark this page or use the **Recent Jobs** menu at the left to access your job's results.

MEME Suite 5.5.7

Queue status: OPEN

Jobs running: 14

Jobs waiting to run: 160

Submitted

Expires

(Primary) Sequences

Motifs

Motif Width

Gene Regulation

Utilities

Manual

Guides & Tutorials

Sample Outputs

File Format Reference

Databases

Download & Install

Help

Alternate Servers

Authors & Citing

Recent Jobs

MEME-ChIP 2:54 PM

Clear All

Previous version 5.5.6

Job Details

Submitted 4/15/2025, 2:56:26 PM

Expires 4/19/2025, 2:56:26 PM

(Primary) Sequences A set of 55392 DNA sequences, between 227 and 46585 in length (average length 854.6), from the file `[K562_POLR2A_peaks.narrowPeak.Sequences.fasta]`.

Motifs A set of 1956 motifs, between 5 and 35 in width (average width 12.5), from the database `[JASPAR CORE (2022)]`.

Motif Width MEME and STREME will find motifs that are between 6 and 15 wide (inclusive).

Gene Regulation A 2-order background model generated from the supplied sequences.

Utilities Parse Genomic Coordinates

Manual Motif sites will be output in genomic coordinates if possible.

Guides & Tutorials Discovery Mode

Sample Outputs Classic: optimizes the E-value of the motif information content.

File Format Reference MEME Site Distribution

File Format Reference MEME will only find motifs with zero or one occurrence (of a contributing motif site) per sequence.

File Format Reference MEME will search for 3 motifs.

Databases STREME p -value Threshold

Databases STREME will find motifs with a p -value ≤ 0.05 .

Databases STREME Site Positional Distribution Plot

Databases Sequences are aligned on their centers.

Databases CentriMo Match Score

Databases CentriMo will only use motif sites if they have a match score ≥ 5 .

Download & Install CentriMo E-value Threshold

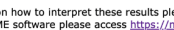
Download & Install CentriMo will only report motifs which have a region enrichment E -value ≤ 10 .

Help CentriMo Local

Help CentriMo will only find central enriched regions.

Alternate Servers CentriMo Store IDs

Alternate Servers CentriMo will store the sequence identifiers and their mapping to the best region of all significant motifs.



MEME-CHIP
Motif Analysis of Large Nucleotide Datasets

For further information on how to interpret these results please access <https://meme-suite.org/meme/doc/meme-chio-output-format.html>.
To get a copy of the MEME software please access <https://meme-suite.org>.

If you use MEME-CHIP in your research, please cite the following paper:
Philip Machanick and Timothy L. Bailey, "MEME-CHIP: motif analysis of large DNA datasets", *Bioinformatics* 27(12):1696-1697, 2011. [\[full text\]](#)

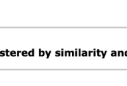
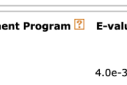
[MOTIFS](#) | [PROGRAMS](#) | [INPUT FILES](#) | [PROGRAM INFORMATION](#) | [SUMMARY IN TSV FORMAT](#) | [MOTIFS IN MEME TEXT FORMAT](#)

MOTIFS

The significant motifs (E-value ≤ 0.05) found by the programs MEME, STREME and CentriMo; clustered by similarity and ordered by E-value.

Expand All Clusters

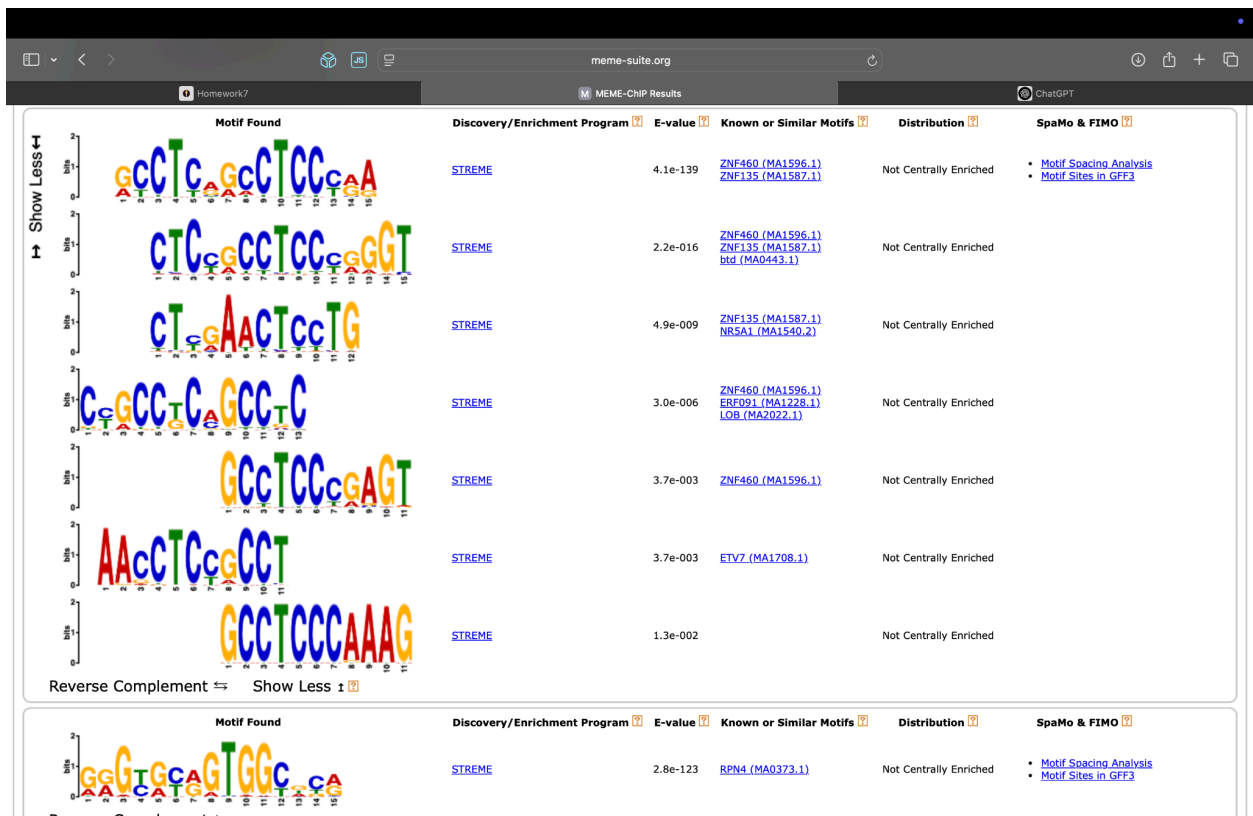
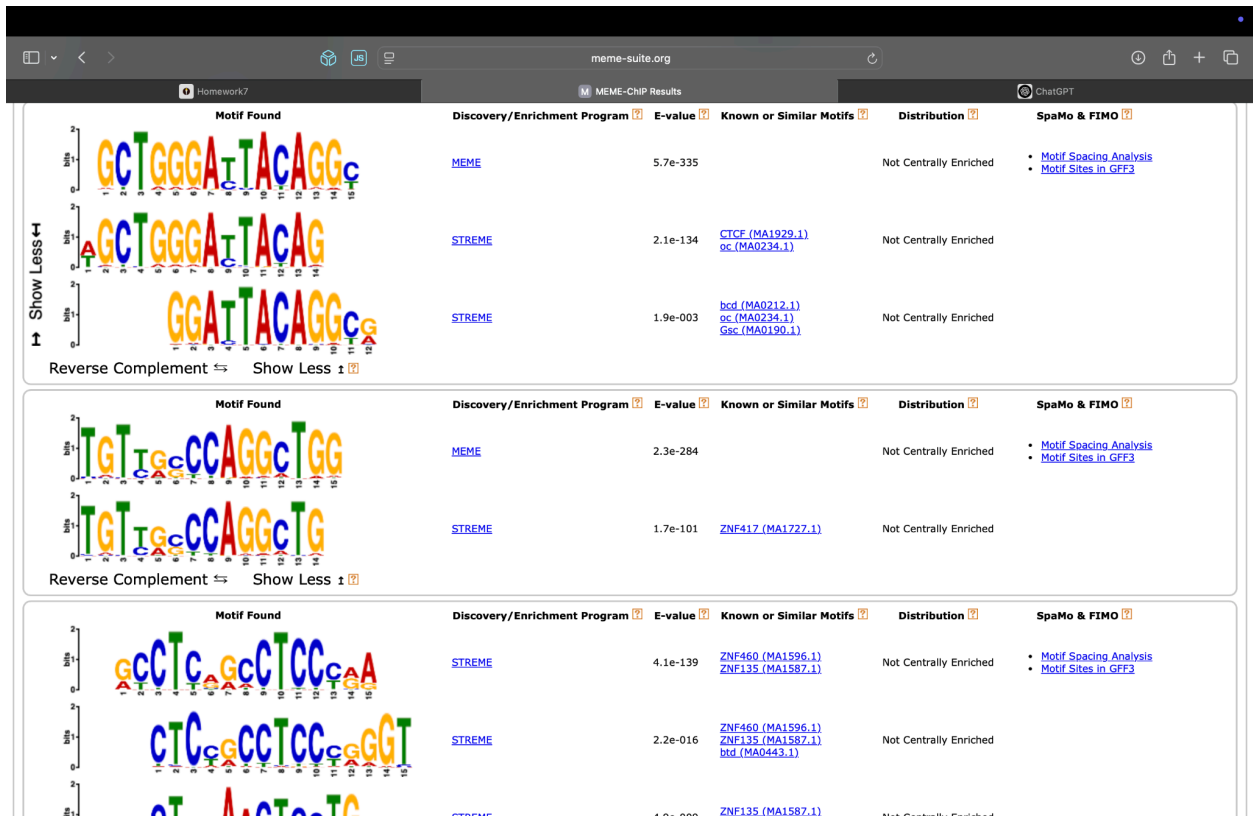
Collapse All Clusters

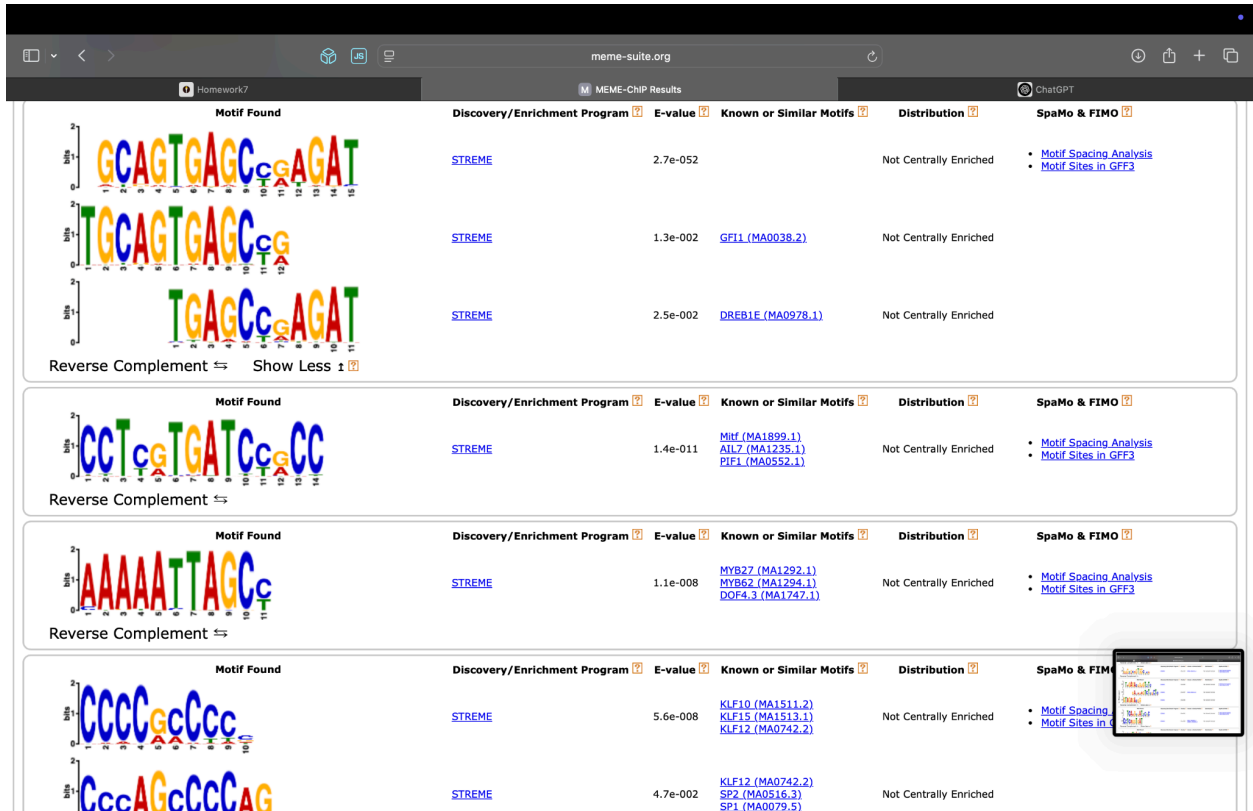
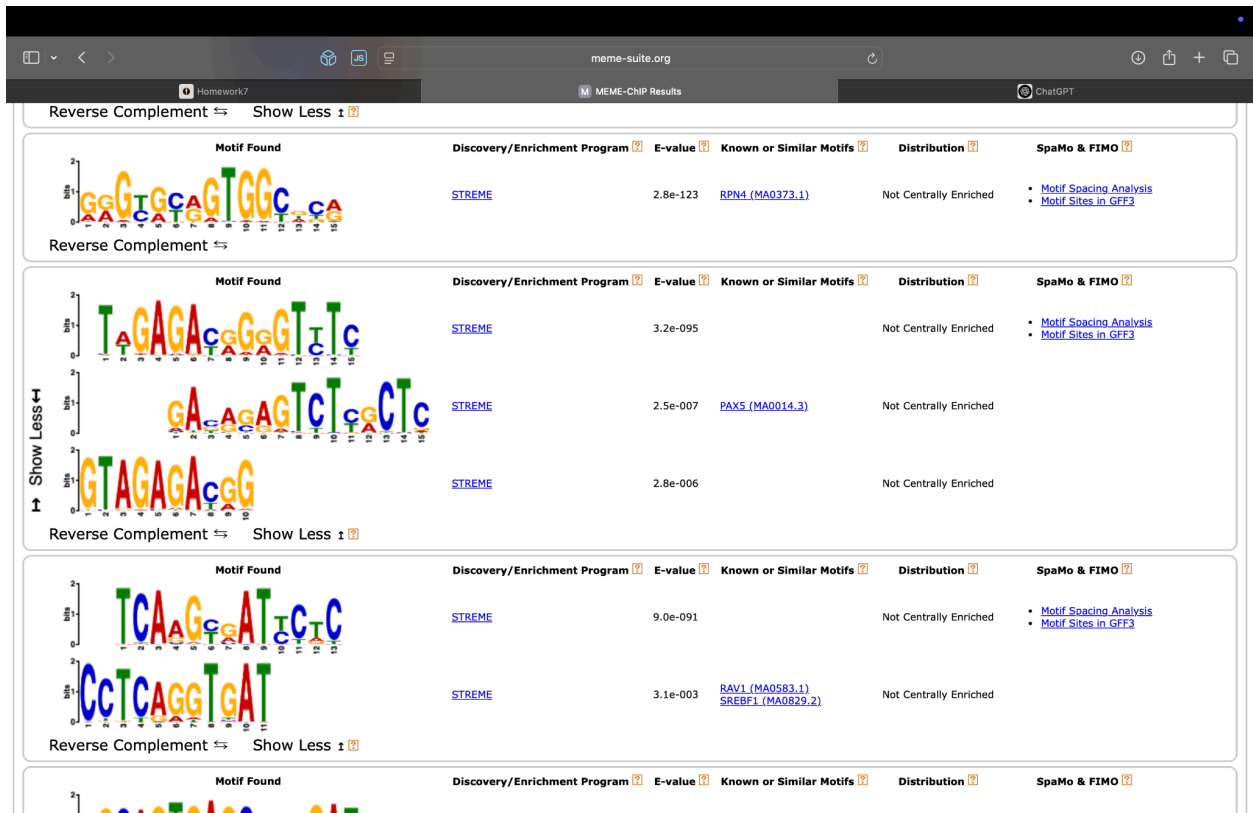
Motif Found	Discovery/Enrichment Program	E-value	Known or Similar Motifs	Distribution	SpaMo & FIMO
	MEME	4.0e-344	FoxM (MA1871.1) DOF3.6 (MA1274.1) FoxE (MA1863.1)	Not Centrally Enriched	Motif Spacing Analysis Motif Sites in GFF3
	STREME	1.6e-053	FoxE (MA1863.1) SGR5 (MA1159.1) FoxM (MA1871.1)	Not Centrally Enriched	
	STREME	2.1e-010	CDF5 (MA1268.1) DOF3.6 (MA1274.1) DOF5.8 (MA1267.1)	Not Centrally Enriched	

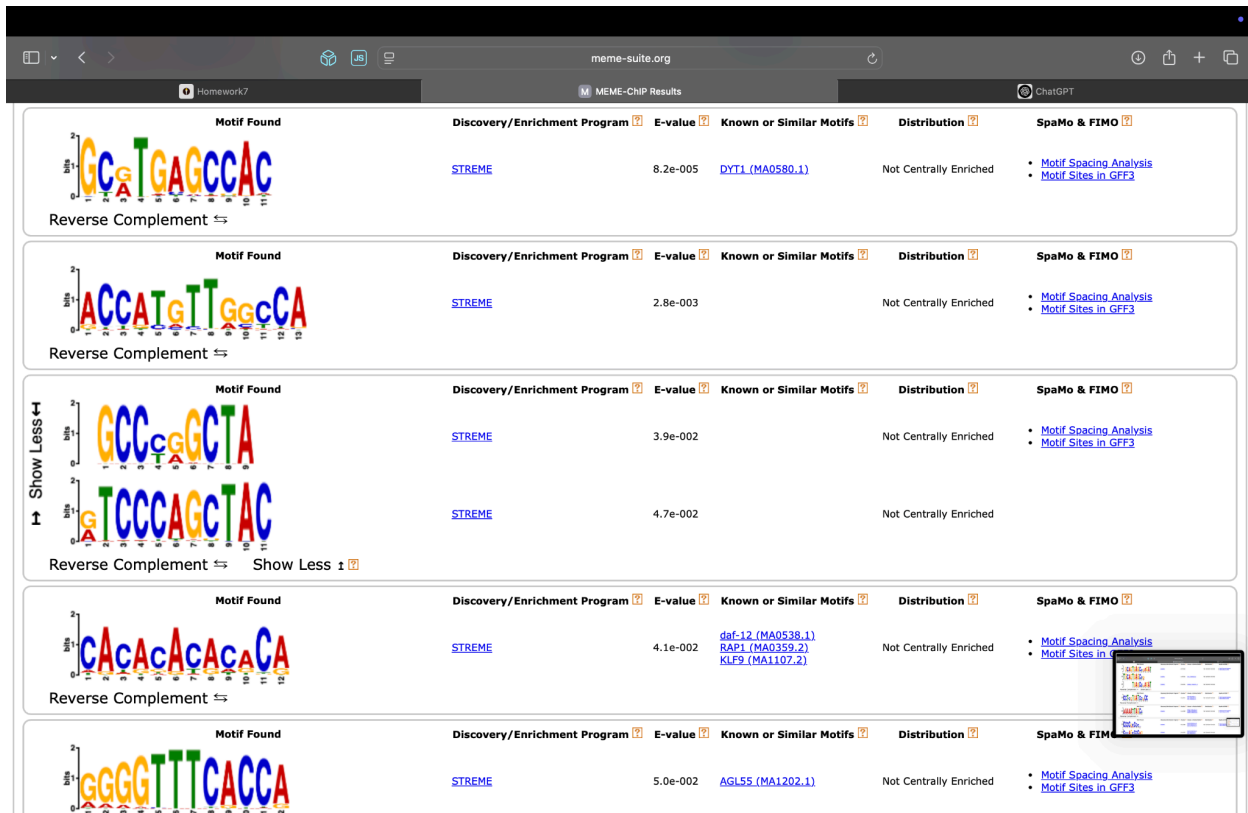
Reverse Complement

Show Less

Motif Found	Discovery/Enrichment Program	E-value	Known or Similar Motifs	Distribution	SpaMo & FIMO
	MEME	5.7e-335		Not Centrally Enriched	Motif Spacing Analysis Motif Sites in GFF3







1. MOTIF 1:

- **Sequence (Consensus):** WWTTTTTCTWTTTTT
- **Type:** MEME
- **Length:** 15
- **Letter-probability matrix:**
 - This motif is heavily enriched for **T** throughout, particularly in the middle and end.
 - The central positions (4–13) show **T** probability exceeding 90%.
 - Low probabilities for **G** and **C**, indicating high specificity.
 - Likely involved in promoter or repressor elements, possibly acting in nucleosome-free regions.

2. MOTIF 2:

- **Sequence:** GCTGGGATTACAGGC
- **Type:** MEME
- **Length:** 15
- **Letter-probability matrix:**

- Strong enrichment for **G** at the first 6 and last 3 positions.
- Positions 7–10 include **ATTAC**, resembling enhancer-related sequences.
- Overall, this is a **G/A-rich motif**, likely bound by **specific transcription factors** (e.g., RUNX or NF-Y).
- Highly conserved with an E-value of **5.7e-335**.

3. MOTIF 3:

- **Sequence:** TGTTGSCCAGGCTGG
- **Type:** MEME
- **Length:** 15
- **Letter-probability matrix:**
 - High **T** content at the 1st, 3rd, and 4th positions.
 - **C and G** dominate in the middle, possibly indicating **GC box** regions.
 - Ends with a strong **GG** bias.
 - Could represent binding of **SP1** or **ELK-family** transcription factors.

4. MOTIF 4:

- **Sequence:** GCCTCRGCCTCCCRA
- **Type:** STREME
- **Length:** 15
- **Letter-probability matrix:**
 - This motif is **GC-rich**, with high probability of **C** at positions 2, 3, 5, 8–13.
 - **Purine enrichment** (A/G) is evident at positions 1 and 14.
 - Suggests a **zinc finger TF binding site**, likely in **SP/KLF** families.

5. MOTIF 5:

- **Sequence:** CTGTAATCCCAGCW
- **Type:** STREME
- **Length:** 14
- **Letter-probability matrix:**
 - Conserved **TAATCCC** core is observed.
 - Tendency toward **C-rich tails**, and **A** in the center.
 - May be associated with enhancer or silencer activity.

6. MOTIF 6:

- **Sequence:** GRGYGYRGTGGCDCR
- **Type:** STREME
- **Length:** 15
- **Letter-probability matrix:**

- Alternating **G/Y (C/T)** pattern, producing a repetitive, symmetric structure.
- Strong **G** enrichment throughout.
- Suggestive of **GC-repeat-binding TFs**, possibly **ZBTB** or **G-rich structural motifs**.

7. MOTIF 7:

- **Sequence:** CAGCCTGGSCAACA
- **Type:** STREME
- **Length:** 14
- **Letter-probability matrix:**
 - Central **GGSCAAC** motif flanked by C and A-rich regions.
 - Likely involved in **gene regulatory regions**, especially near **splicing sites or intron-exon boundaries**.

8. MOTIF 8:

- **Sequence:** TWGAGAYRGRGTYTC
- **Type:** STREME
- **Length:** 15
- **Letter-probability matrix:**
 - Very mixed sequence with **TGAG**, **GRG**, and **TYT** repeats.
 - Likely recognized by **dimeric transcription factors** like **NF-kB** or **bZIP** family.
 - High complexity suggests strong regulatory function.

9. MOTIF 9:

- **Sequence:** TCAAGYGATYCTC
- **Type:** STREME
- **Length:** 13
- **Letter-probability matrix:**
 - Central **YGATYC** core is highly conserved.
 - Tends to occur in immune-related response elements like **ISRE** or **IRF** sites.
 - Balanced usage of **A/T** and **C/G**.

10. MOTIF 10:

- **Sequence:** ATTTTGTATTTTA
- **Type:** STREME
- **Length:** 15
- **Letter-probability matrix:**
 - Strong **poly-T track** with central **TGTAT** pattern.
 - May mimic **TATA boxes** or poly-A/T tracts common in **core promoters**.
 - Suggests involvement in **transcription start regions**.